1. Reduced-round neural distinguishers under crypto-ml-v1 (2026-11-02)

Study: `research/neural_distinguishers_20261102` · **protocol frozen:** `d7e1cc432` · **run:** task cards `CARD-neural-distinguishers-20261102-1` (interrupted, 33/35 arms) and `-2` (recovery of cell aes-r4, completed; receipt `b8902ba6…`) at commit `f0993f828` · **rows:** `results/rows.jsonl` (sha256 `ab49ea47…`) · **summary:** `results/summary.json` (`2405a91a…`) · **host:** session container, 4 vCPU, CPU only, $0.

1.1. What was built

`docs/crypto-ml-v1.md` defined the lab's ML-cryptanalysis contract with no implementation. This study implements it end to end in native Rust (no Python): Speck32/64 and AES-128 generators checked against published vectors (Speck paper Appendix C; FIPS-197), Gohr's real-vs-random task with exactly balanced classes and one fresh key per sample (key-disjoint splits by construction), a 128-128 MLP and logistic regression trained from scratch, a classical difference-table naive-Bayes baseline, and per-cell positive, negative (labels permuted in train and test) and A/A controls, with a Bonferroni-corrected decision rule over 21 tests at $\alpha = 0.01$.

1.2. Results (measured; held-out accuracy, n_test = 131 072 per cell, 95% Wilson intervals)

cell	MLP	logistic regression	difference-table NB	neg. control	A/A Δ
Speck r3 (pos. ctrl)	0.9993 [0.9991, 0.9994]	0.9551	0.9656	0.5000	+0.0001
Speck r5	**0.8775** [0.8758, 0.8793]	0.7517	0.7502	0.5018	−0.0001
Speck r6	**0.6938** [0.6913, 0.6963]	0.5836	0.5750	0.5004	−0.0010
Speck r7	**0.5190** [0.5163, 0.5217]	0.5008 (n.s.)	0.5001 (n.s.)	0.5022	+0.0012
AES r2 (pos. ctrl)	1.0000	1.0000	1.0000	0.4996	+0.0000
AES r3	0.4998 (n.s.)	0.5046 (n.s.)	**0.5221** [0.5194, 0.5248]	0.4988	+0.0011
AES r4	0.5016 (n.s.)	0.5015 (n.s.)	0.5016 (n.s.)	0.4989	−0.0012

"n.s." = the Bonferroni-corrected Wilson lower bound is ≤ 0.5: *no distinguisher found at this size*. Every other entry *distinguishes* under the frozen rule. All seven cells are valid: positive controls detected (> 0.9), every negative control's 95% interval contains 0.5.

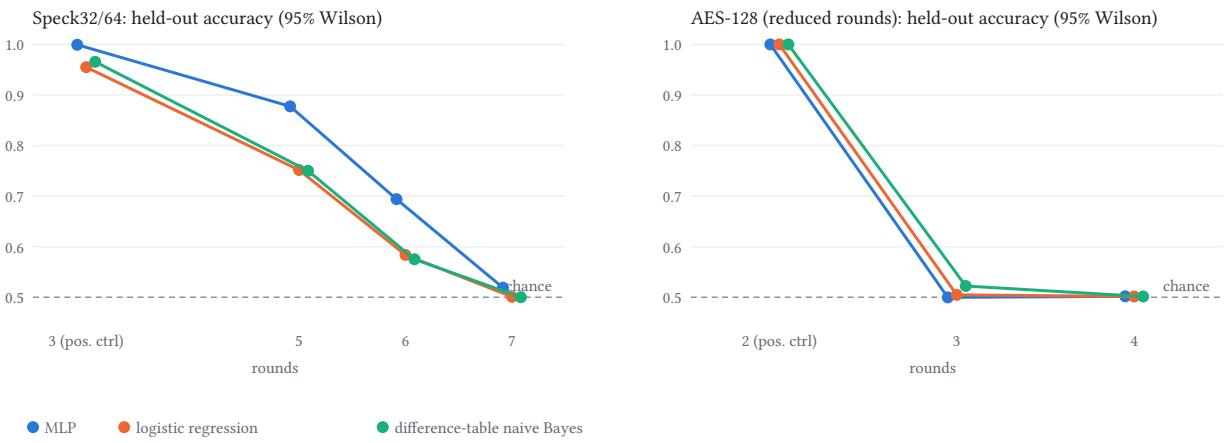


Figure 1: Held-out accuracy by round count, one panel per cipher, three models, 95% Wilson bars; dashed line = chance. Recorded values from `results/summary.json`; cards −1/-2.

1.3. Reading the results (comparative claims, at the protocol's scope)

- **Speck32/64, 5–7 rounds.** The MLP distinguishes at 5, 6 and 7 rounds and its 95% interval lies above both classical baselines at every Speck round count. At 7 rounds only the MLP distinguishes (corrected lower bound 0.5142). Advantage over chance: 0.378 / 0.194 / 0.019 at 5 / 6 / 7 rounds, against an A/A seed spread ≤ 0.0012. For orientation only (not reproduced, not used by the rule): Gohr's ResNet with 10^7 samples reports 0.929 / 0.788 / 0.616; this MLP with 2^{20} samples reaches 0.878 / 0.694 / 0.519.
- **AES-128, 3 rounds.** The classical difference-table baseline distinguishes (0.5221, lower bound 0.5172) while the bit-level MLP and logistic regression do not. A small MLP on raw bits is therefore **not** uniformly stronger than a classical statistic: here the byte-level difference distribution carries signal the MLP did not learn at this size.
- **AES-128, 4 rounds.** No model distinguishes at this size.

1.4. Execution record

Card −1 ran all cells in order and was killed by the agent harness's 30-minute limit on background shell commands (not by a card timeout) after 33 of 35 arms; no receipt was written. Card −2 re-ran only cell aes-r4 with the same frozen code and seeds and appended its 5 rows. Determinism check: the three aes-r4 arms present in both runs agree exactly on `correct` (65 746, 65 588, 65 737). The summary uses the last row per (cell, model, arm). The experiment-director's assessment (`cards/ASSESSMENT-20261102.md`) classifies card −2 *completed* (all digests recomputed and matching) and card −1 *interrupted*. The relay check (`task_card.py check-relay`) is **inconclusive**: it reports one expected receipt line missing although that line is present verbatim in the tool's own output, and the report file it was run on was assembled from the tool output rather than the executor's message, so it does not independently verify the relay.

1.5. Scope and limits

Distinguisher accuracy is neither key recovery nor a full-round result. The difference-table baseline is weaker than Gohr's full Markov DDT distinguisher, which is not implemented, so "beats the classical baselines" means these baselines. One architecture, one sample size and one epoch budget per cell; a larger model or more data may distinguish AES r3/r4 or improve Speck r7. Training times are recorded per row but are not a performance claim. The EKS route of AGENTS.md §9a was unavailable (no AWS identity); everything ran in the session container. No independent `review-run` record exists yet.

1.6. Next technical steps

1. Add Gohr's Markov DDT baseline for Speck r5–r7 (the stronger classical reference).
2. AES r3: a byte-level model (embedding per byte difference) to test whether the MLP's miss is representational.
3. Reuse this training/evaluation harness for the ECDLP dataset models (`docs/ml-dataset/training-protocol-20261101.md`) on development data only.